

RS003 Extension — Design Your Game Art with AI

Topic: Prompting Google Gemini for 32-bit Game Sprites · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This is a bonus mission. Your job this week is **not coding** — it is being the **art director**. So far your game uses shapes you drew with code. Now you write careful instructions (a **prompt**) and Google Gemini draws **32-bit style** sprites for your game — 3 enemy ships, 1 player ship, and 1 background. A good prompt is like a good recipe: the clearer you are, the better the picture.

 Words to know: **prompt, sprite, 32-bit, transparent background, asset**

 Open Gemini together with a grown-up or your teacher. Always check the picture before you use it.

1 · What Makes a Good Prompt?

A computer artist cannot read your mind. It only knows what you **say**. A strong sprite prompt names **six things**:

Part	Question it answers	Example
Subject	What is it?	a tiny alien fighter ship
Style	What art look?	32-bit pixel art, retro arcade
View	From what angle?	top-down view, seen from above
Colours	What palette?	purple and green, glowing
Background	What is behind it?	plain transparent background, no scenery
Use	Where will it go?	a game sprite for a 2D shooter

Read this prompt. Underline the six parts above inside it:

"A tiny alien fighter ship, 32-bit pixel art, retro arcade style, top-down view seen from above, purple and green with glowing lights, on a plain transparent background, as a game sprite for a 2D shooter."

Why do we ask for a *transparent background*? (Hint: think about putting the ship on top of your own sky.)

2 · The Prompt Recipe 🍷

Here is your fill-in-the-blank recipe. Copy it, swap the blanks for your own ideas, and you have a prompt. The words in **bold** stay the same every time — they are what make it look like a game sprite.

```
A _____ (what it is),  
in 32-bit pixel art, retro arcade style,  
top-down view seen from above,  
_____ colours (your palette),  
on a plain transparent background, no scenery,  
as a single game sprite for a 2D Pygame shooter.
```

Fill the recipe for ONE idea right now — your player ship:

```
A _____  
in 32-bit pixel art, retro arcade style, top-down view seen from above,  
_____ colours,  
on a plain transparent background, no scenery, as a single game sprite.
```

3 · Spot the Weak Prompt 🐉

Each prompt below would give a messy result. Say what is missing and rewrite it using the recipe.

Prompt A — *"a spaceship"*

What is missing?

Your better prompt:

Prompt B — *"a cool enemy ship with an amazing detailed realistic 3D background and explosions and a planet and stars everywhere"*

Hint: you only want the ship by itself. A busy background is hard to drop into your game.

Your better prompt:

Prompt C — *"make 5 ships"*

Hint: ask for **one** sprite at a time so each picture is clean and the right size.

Your better prompt:

4 · Design Your Five Assets

Plan all five **before** you open Gemini. Give each enemy its own personality through colour and shape words (spiky, round, heavy, fast).

Player ship

A ____, in **32-bit pixel art, top-down view**, ____ colours, transparent background, single game sprite.

Enemy ship 1

Enemy ship 2

Enemy ship 3

Background

The background is the one asset that should **fill the whole picture** (no transparent part). Name a mood and a few details.

Hint: "a deep space night sky background, 32-bit pixel art, dark blue with tiny stars and a far away planet, wide 2D game background, top-down view, no ships."

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5 • Make It 📷

 Generate and save all five assets

1. Open Google Gemini with your teacher or a grown-up.
2. Paste one prompt at a time. If the picture is wrong, **change one word** and try again — that is real prompt engineering.
3. Save each finished picture (for example `ship.png` , `enemy1.png` , `enemy2.png` , `enemy3.png` , `background.png`).
4. You should end with **5 images**: 1 player ship, 3 enemy ships, and 1 background.

💡 No coding this week. Your job is the art only — in Week 16 you swap these pictures into the game with `pygame.image.load(...)`.

Send all **5 saved pictures**. Then explain what you did — write 4 to 6 sentences. Use these starters:

The prompt I was most proud of was ...

I had to change a word when ...

I asked for a transparent background because ...

The enemy ship I made look the scariest is the one that ...

One thing the AI got wrong was ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Record a short video of **yourself** talking through your art. You are not showing a game — you are explaining your **prompts** and **why** you chose the words you did. Try to use these words: **prompt, sprite, 32-bit, transparent background, asset**.

1. Show your 5 pictures one at a time.
2. Read one prompt out loud and point to each of the six parts (subject, style, view, colours, background, use).
3. Explain **why** you chose those colours and words for that ship.
4. Show one picture the AI got wrong and the word you changed to fix it.

Write what you will say in your video. Plan it here before you record.

Submit 

Send your **5 pictures** + your walkthrough video to teacher on **KakaoTalk**.